LINUX SYSTEM ADMINISTRATION

UFW Firewall Configuration and Traffic Control

Practical Assignment — Questions with Answers

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Topics: UFW installation • Default policies • SSH access • Outbound HTTP/HTTPS • Rule listing • Local web-server impact

Safety note: Setting default outbound traffic to deny can interrupt DNS, package installation, updates, and other connections. Keep console/VM access available and add any required outbound rules before enabling this policy. Commands below are a sample procedure; verify on your own system.

Objective

Install and enable UFW, permit incoming SSH on TCP port 22, deny other incoming traffic, allow outbound HTTP/HTTPS only, save the current rules to firewall_rules.txt, and explain the effect on a local web development server.

Important platform note

UFW is common on Ubuntu/Debian. On Red Hat Enterprise Linux, firewalld is normally the default firewall manager and UFW may not be available in the standard repositories. These commands are intended for a Linux system where UFW is supported. Do not mix firewall managers without understanding the consequences.

Question 1 — Install, enable, and verify UFW

Q. How do you install UFW if needed, enable it, and check its status?

Answer / Commands (Ubuntu/Debian)

```
sudo apt update
sudo apt install -y ufw
sudo ufw status verbose
```

Before enabling, set the required policies and rules in Questions 2 and 3. Then enable UFW. This avoids enabling a restrictive firewall before the intended allow rules are added.

Question 2 — Allow incoming SSH and block other incoming traffic

Q. How do you allow incoming SSH only on port 22 and block all other incoming connections?

Answer / Commands

```
sudo ufw default deny incoming
sudo ufw allow in 22/tcp
sudo ufw status numbered
```

The default incoming policy denies unsolicited inbound traffic; the explicit rule allows TCP port 22 for SSH. If SSH uses a different port, adjust the rule first. If connected remotely, confirm port 22 is correct before enabling the firewall so you do not lock yourself out.

Question 3 — Allow outbound HTTP/HTTPS and deny other outbound traffic

Q. How do you allow outgoing connections on ports 80 and 443 while blocking all other outbound connections?

Answer / Commands

```
sudo ufw allow out 80/tcp
sudo ufw allow out 443/tcp
sudo ufw default deny outgoing
sudo ufw enable
sudo ufw status verbose
```

Order matters: add the specific outbound allow rules before applying the default deny outgoing policy. This policy is intentionally restrictive. DNS (usually UDP/TCP 53), NTP, package repositories, APIs, and other destinations may stop working unless separately allowed. Add only the additional outbound rules your system needs.

Question 4 — List active rules and save output

Q. How do you list current UFW rules and save the output to firewall_rules.txt?

Answer / Commands

```
sudo ufw status numbered | tee firewall_rules.txt
sudo ufw status verbose | tee -a firewall_rules.txt
cat firewall_rules.txt
```

These commands create firewall_rules.txt in the current working directory and save the numbered rules plus verbose status. If you need the file in a specific folder, use that folder's full path. The saved file will contain the output from your system when you run the commands; this PDF does not claim those commands have already been executed.

Question 5 — Impact on a local web server (3–4 lines)

Q. How would this UFW configuration affect a local web server, such as a development server for a Zomato-style food delivery app?

Answer (4 lines)

```
The default incoming-deny policy blocks other devices from reaching the development server unless its port is exp
Incoming SSH on TCP 22 remains allowed, while inbound HTTP/HTTPS ports 80/443 are not allowed by these rules.
Therefore, a web server may work from the same machine but will not be reachable from other devices until the req
Denying other outbound traffic can also break API calls, DNS lookups, database connections, and package downloads
```

Optional: If you intentionally want other devices to access a web server on ports 80 and 443, add these rules only if appropriate for your environment:

```
sudo ufw allow in 80/tcp
sudo ufw allow in 443/tcp
```

Final verification checklist

```
sudo ufw status verbose
sudo ufw status numbered
cat firewall_rules.txt
```

Expected policy summary: incoming default deny; outgoing default deny; explicit incoming TCP/22 allow; explicit outgoing TCP/80 and TCP/443 allow. Verify the actual output on your own system, and capture real screenshots for submission.

Submission reminder

Add your actual Linux distribution/version, date, and genuine terminal screenshots or outputs. Never label sample or expected output as a real execution result.

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